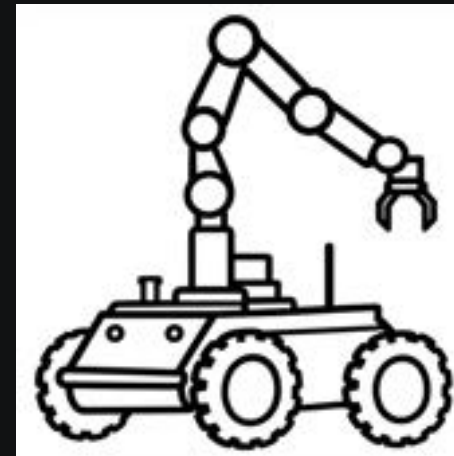
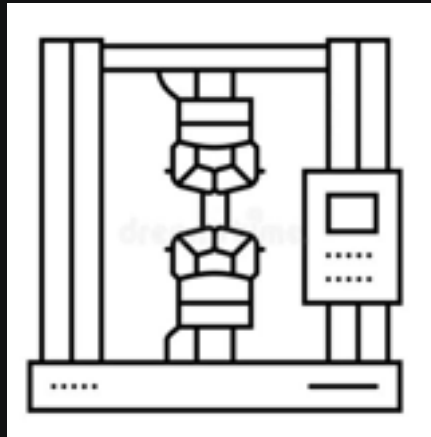
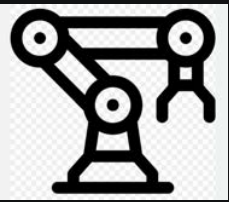


Portfolio by

Betool



Q-Arm Robotic 4 DOF Arm



About

Current robotic arm used for research, education pick-and-place operations is costly, unreliable and lacked the open framework needed for customization and expandability in academic settings causing bottlenecks in research and testing.

A high-precision robotic system was required to perform the complex tasks set by the faculty.

Task

To troubleshoot, re-engineer a 4 DOF robotic arm capable of advanced kinematics calculations, precise positioning within $\pm 2\text{mm}$ accuracy, and seamless integration with existing manufacturing workflows. The system needed to handle payloads up to 2kg while maintaining repeatability standards.

Responsibilities

Implemented forward and inverse kinematics algorithms using MATLAB and Python, designed custom servo motor control systems with PID feedback loops, developed CAD models in SolidWorks with stress analysis simulations, and programmed real-time motion planning algorithms for smooth trajectory execution. Also redesigned the grippers and created a modular platform for developers.

Result

Achieved $\pm 1.5\text{mm}$ positioning accuracy, 60% cost reduction compared to commercial alternatives, successful implementation and scaling the design to 30+ robotic arms and demonstrated 99.2% operational uptime over 3-month testing period with comprehensive documentation for future scaling.

My greatest accomplishment for this project was diagnosing and resolving current spikes in the gripper during shoulder joint movement. I reverse-engineered the power distribution, isolated shared paths, and redesigned the system to consistently reduce current draw from 400–700 mA to $\sim 130\text{ mA}$. I also developed a predictive failure tool to proactively flag overload conditions, helping prevent damage to the QArm with an estimated savings of \$4,500 per unit.

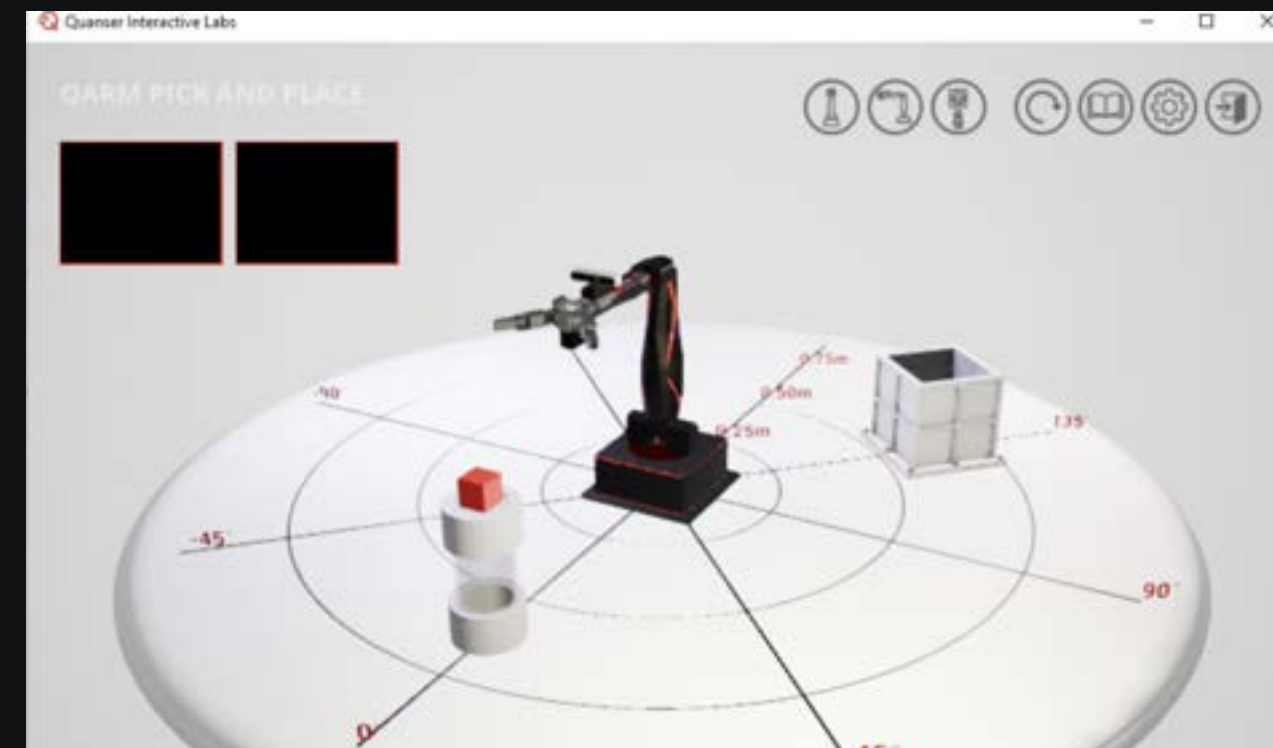
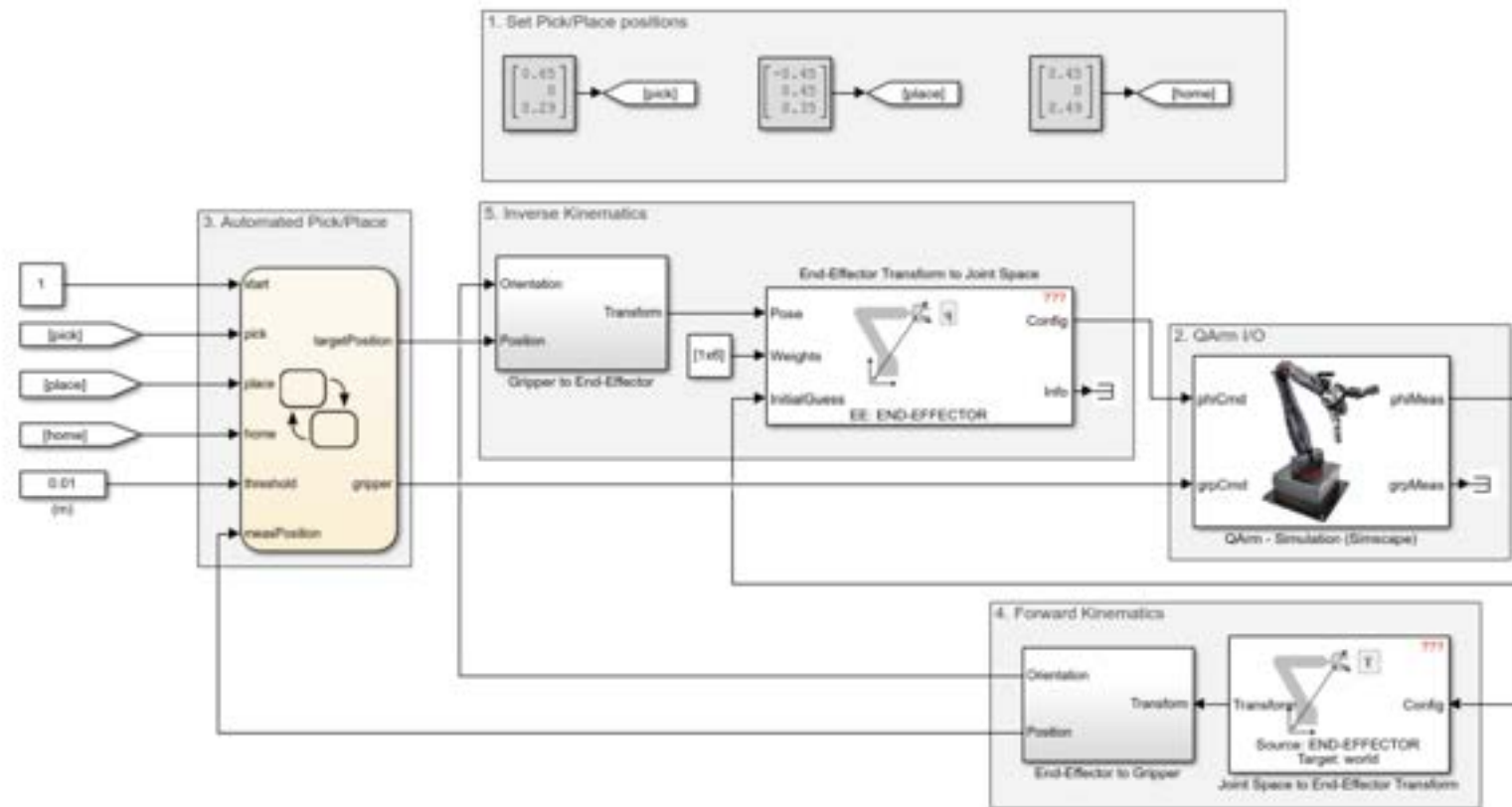
Skills

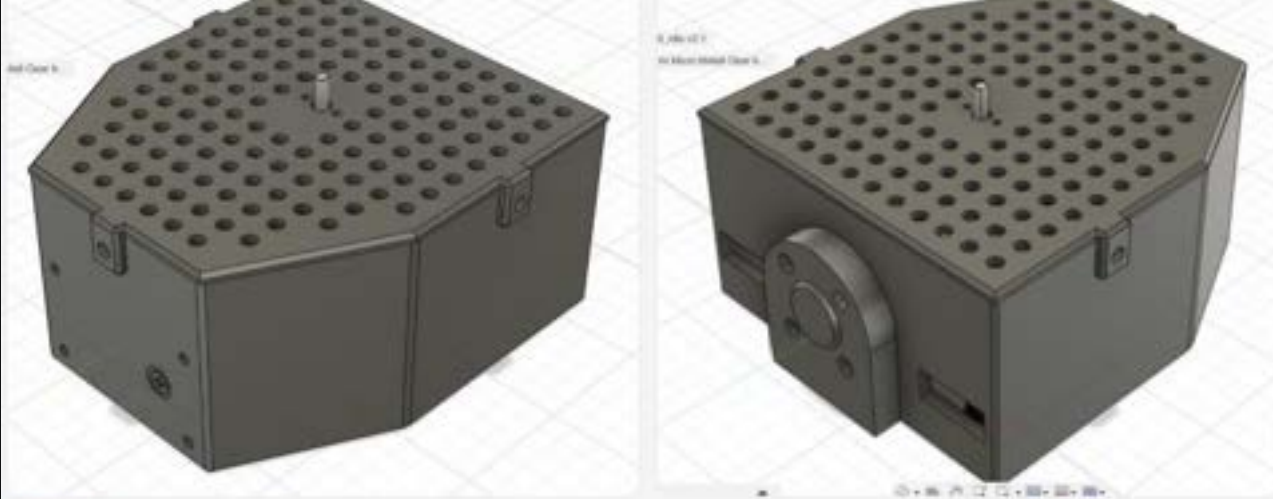
- 3D Printing
- Python
- SolidWorks
- GD&T
- Raspberry Pi
- Motor and Actuator Testing and Implementation
- Worm and Spur Gear Design
- Failure Mode Analysis
- Rapid Prototyping

Dissecting and Testing the Q-Arm



Performing Physical and Software Testing



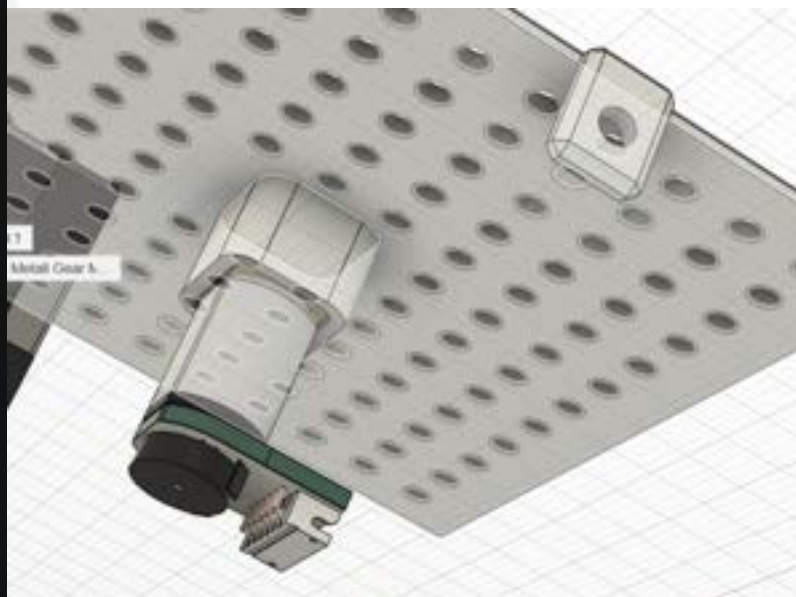


```
restart:

m_p := 0.03; # kg, mass of payload
g := -9.81; # m/s^2
μ_s := 0.35; # static friction, just a guess for PLA
τ_m := 0.1176798; # N*m, torque output of motor
l_f := 0.09; # m, length of finger

F_N := τ_m / l_f; # N, Normal force applied by gripper
F_z_net := m_p * g + μ_s * F_N; # N, Net force in z-direction experienced by payload at max friction

m_p := 0.03
g := -9.81
μ_s := 0.35
τ_m := 0.1176798
l_f := 0.09
F_N := 1.307553333
F_z_net := 0.1633436666
```



I Designed a modular plate for developers

to test their designs and conducted calculations to select an optimal motor

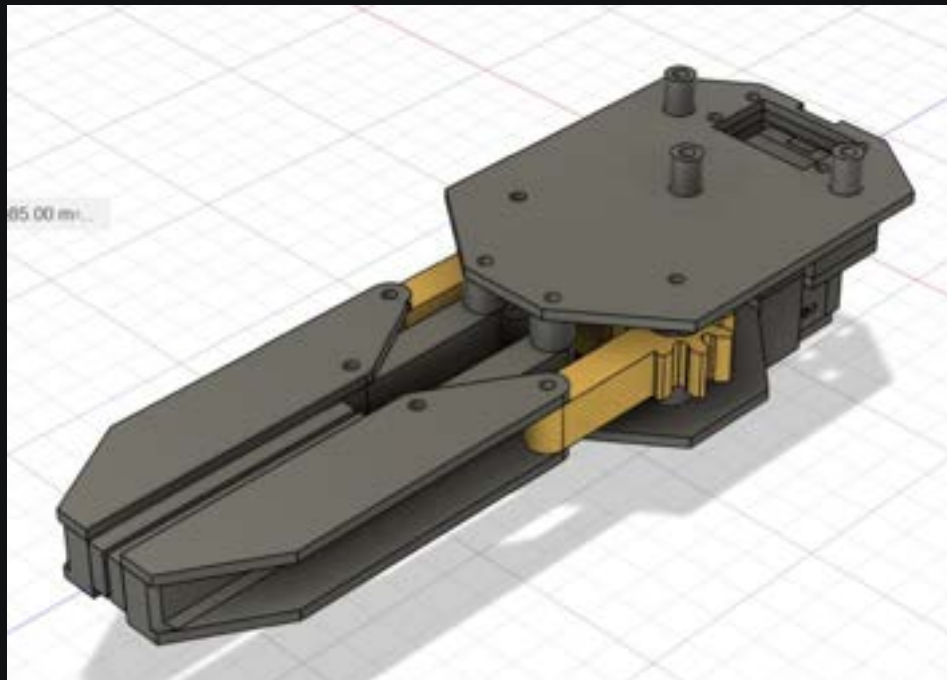
Different Iterations for Gripper Design



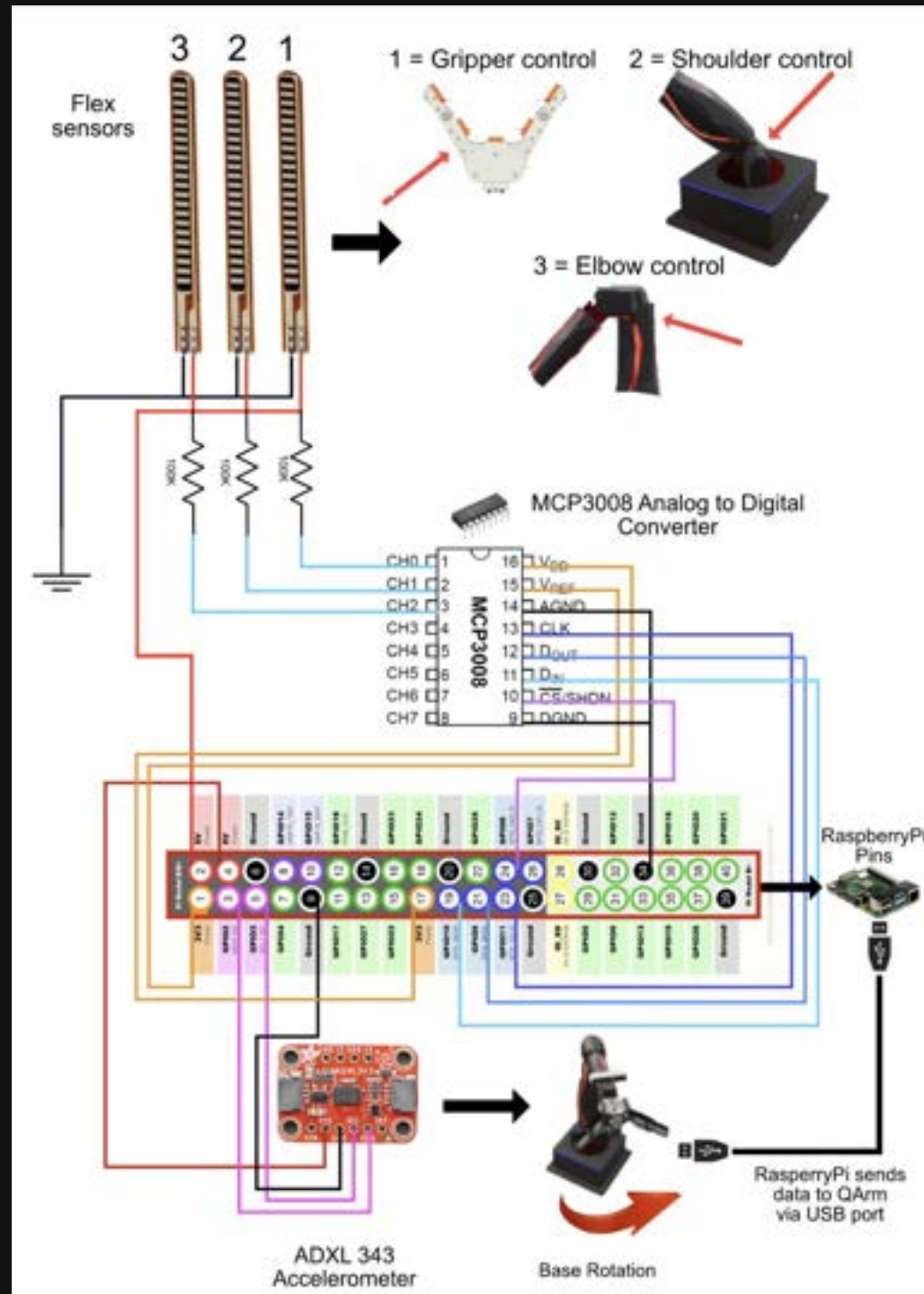
Worm & Spur Gear



Rack and Pinion



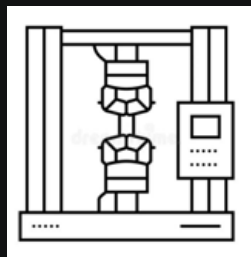
Designed Lab Schematics for Teleoperation



The Final Modular Design of the Arm



Custom Tensile Tester: Material Property Innovation



About

Standard tensile testing equipment costs exceeded \$50,000, making material property testing inaccessible for research labs and small manufacturing companies. Existing solutions lacked customization options for specialized materials and testing protocols required for emerging composite materials.

Task

Develop a custom tensile testing machine capable of applying controlled loads up to 10 kN, measuring strain with high precision, generating standardized test reports, and accommodating various specimen geometries while maintaining ASTM D638 compliance standards.

Responsibilities

Designed mechanical load frame using finite element analysis, integrated high-precision load cells and extensometers, developed LabVIEW control software with real-time data acquisition, implemented automated specimen gripping systems, and created comprehensive calibration protocols ensuring measurement traceability.

Rigorous testing protocols, quality assurance procedures, documentation standards, and performance verification against original requirements and industry benchmarks.

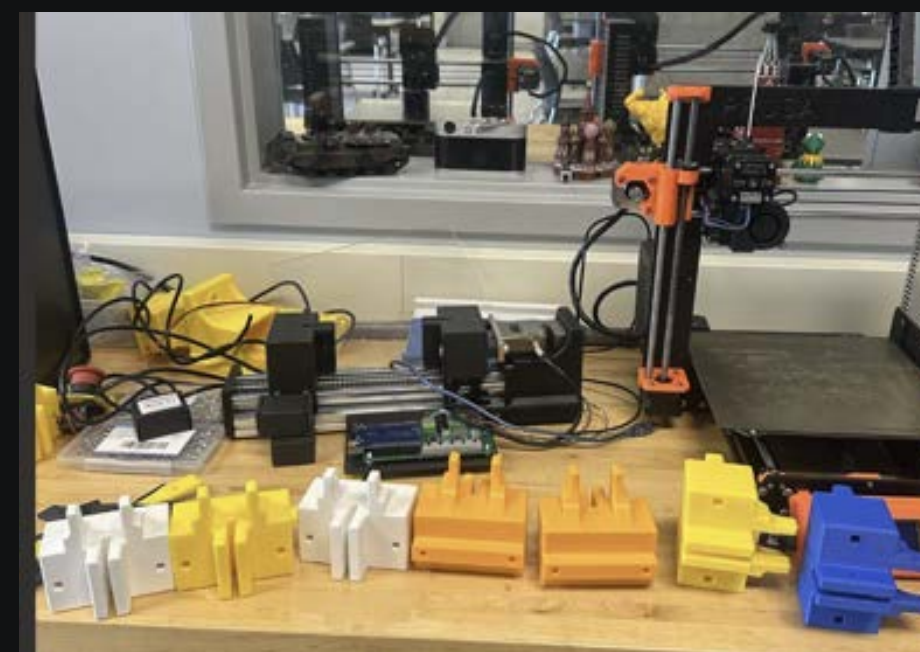
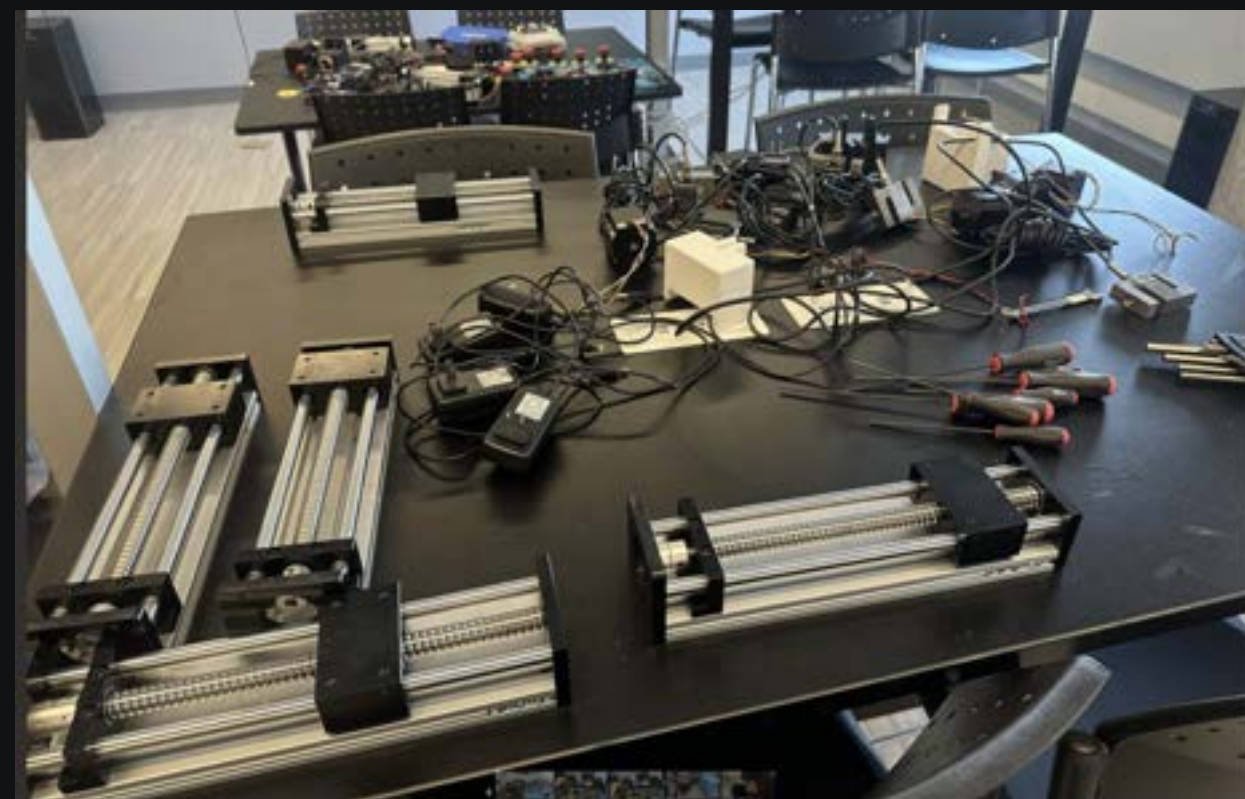
Result Impact

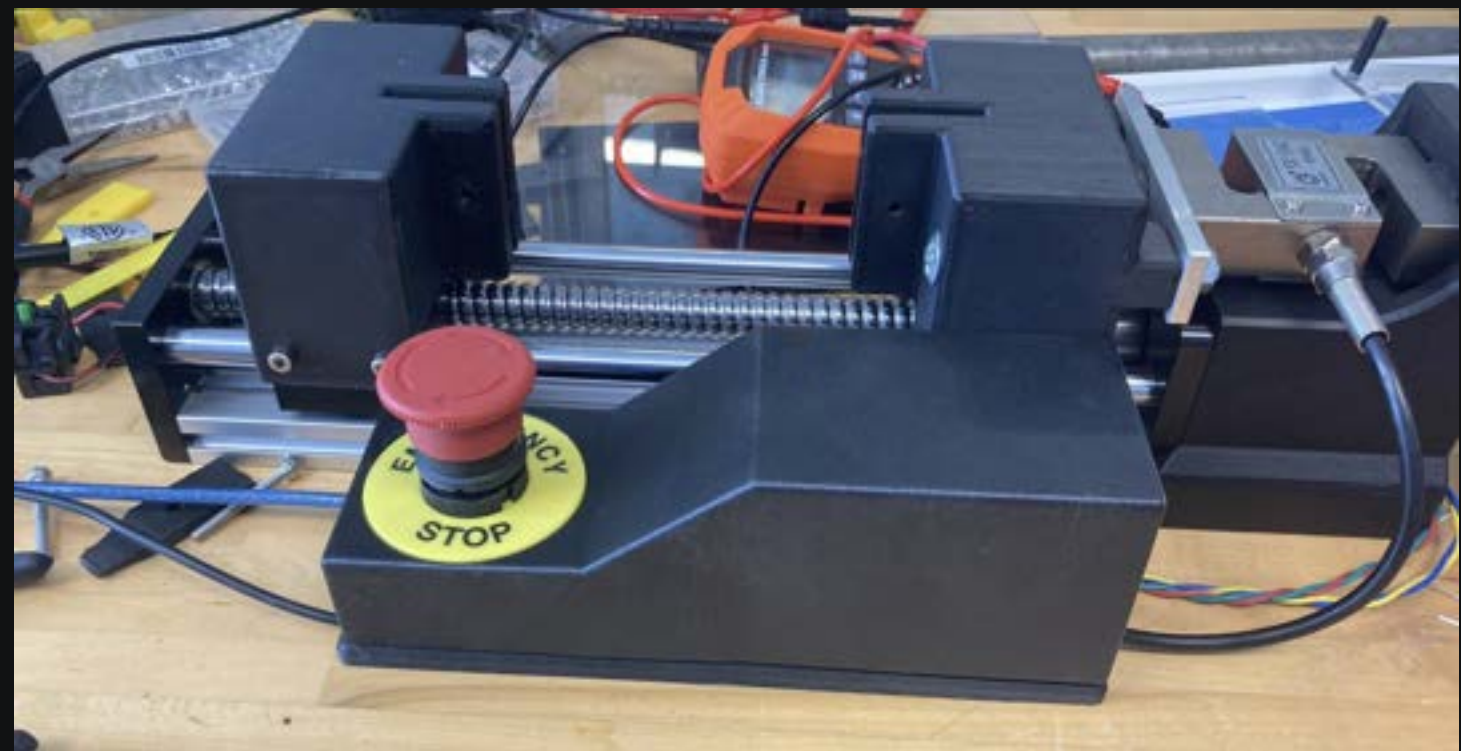
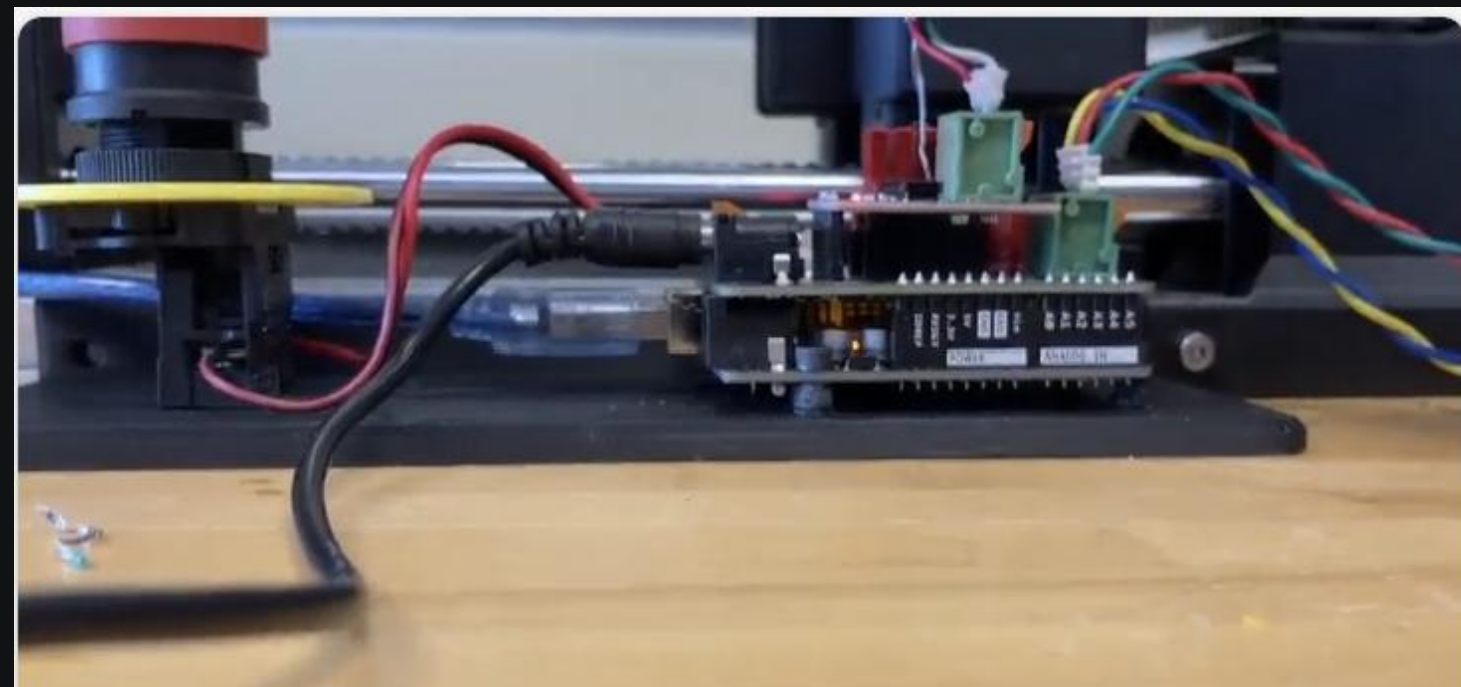
Delivered 85% cost savings compared to commercial systems, achieved $\pm 0.1\%$ load accuracy and $\pm 0.01\text{ mm}$ displacement resolution, enabled testing of 15 different material types, and provided data that is supporting published research papers on composite material properties.

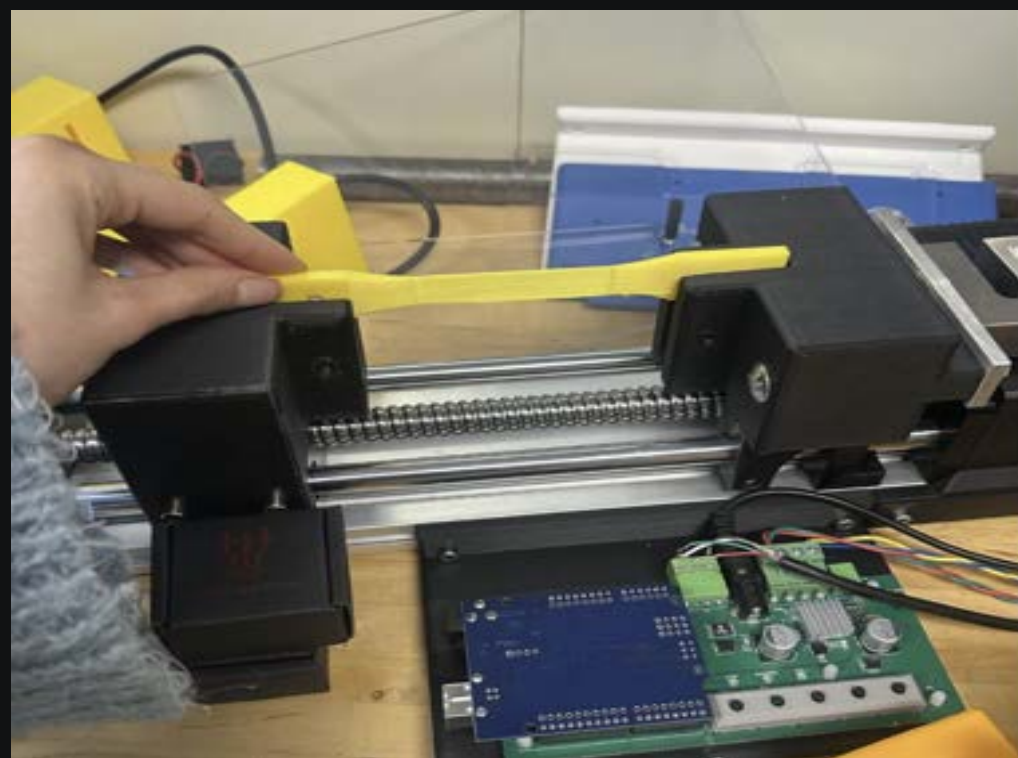
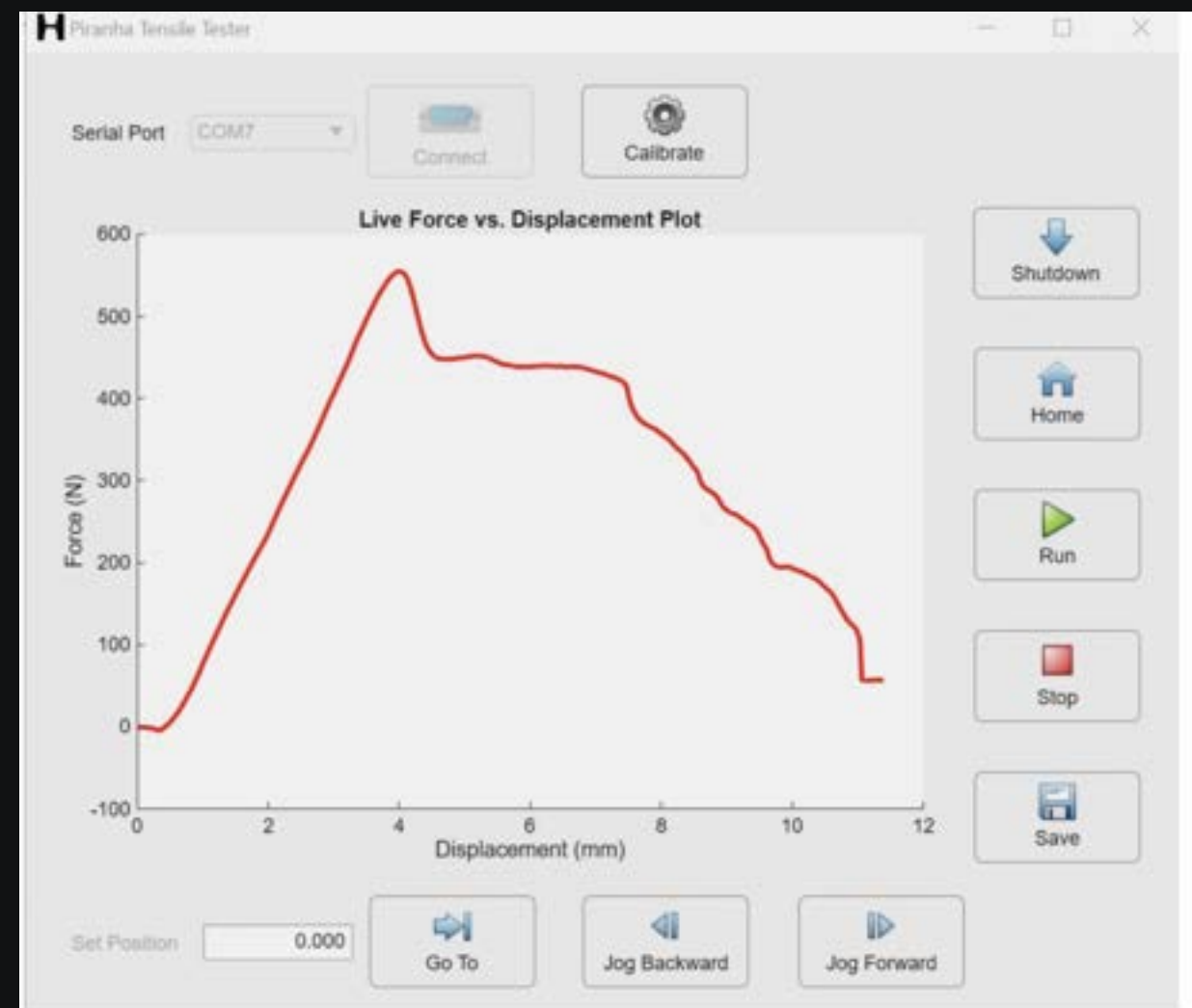
Key Technical Achievement: Successfully validated against certified reference materials, demonstrating measurement uncertainty within ISO 17025 requirements for mechanical testing laboratories.

Skills

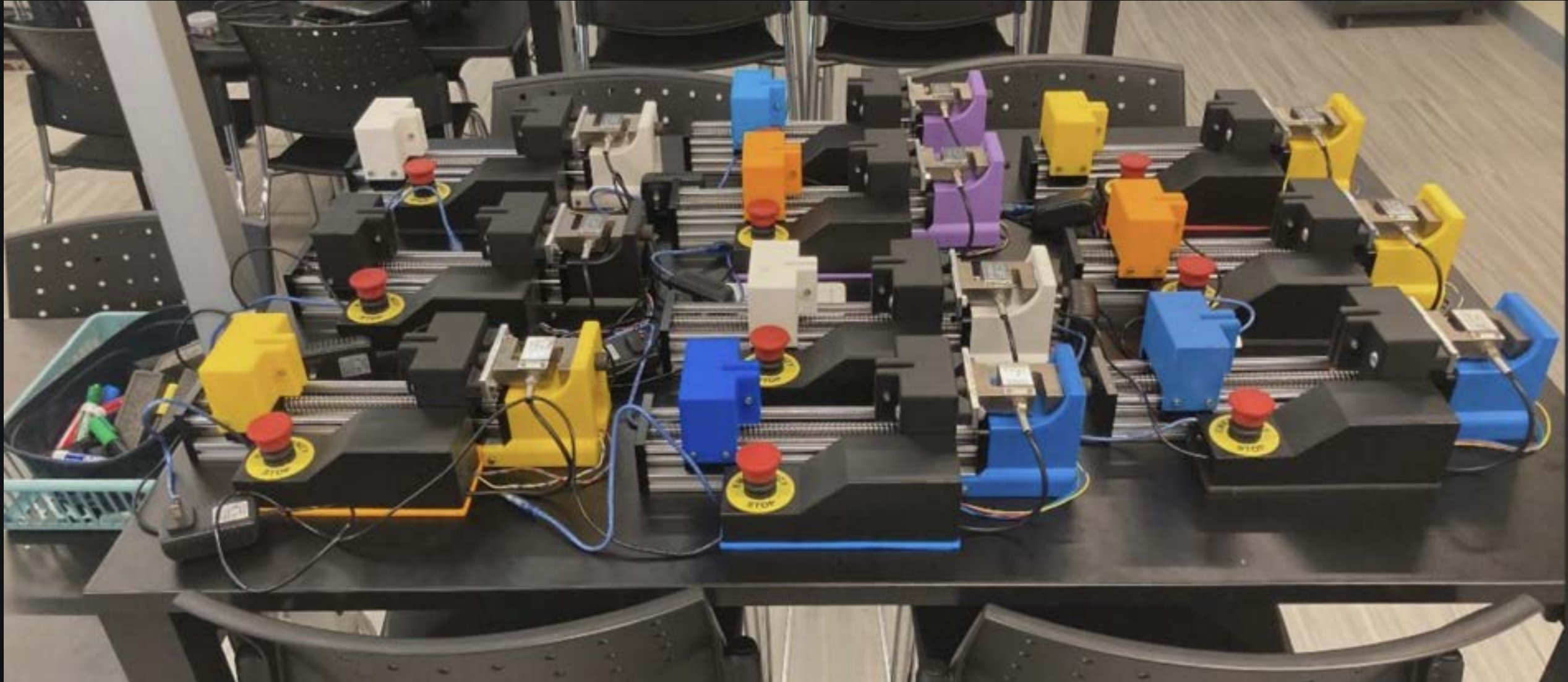
- 3D Printing
- MATLAB
- KiCad
- SolidWorks
- GD&T
- Raspberry Pi
- Motor and Stepper Testing and Implementation
- Failure Mode Analysis
- Rapid Prototyping







Final Design Scaled Testers to 40+ Sets



Octet-Lattice Structure: Helmet Design Revolution



About

Traditional helmet designs struggled to balance protection, weight, and comfort. Conventional foam padding systems provided limited impact distribution and poor ventilation, while meeting safety standards required bulky, uncomfortable designs that have negative impact on the environment.

Task

Engineer an innovative helmet structure using octet-lattice geometry to optimize strength-to-weight ratio, improve impact energy absorption, enhance ventilation, and maintain compliance with DOT and SNELL safety standards while reducing overall helmet weight by minimum 25%.

Responsibilities

Utilized topology optimization algorithms to design octet-lattice internal structure, performed comprehensive FEA simulations for impact scenarios, prototyped using selective laser sintering with titanium alloy, conducted drop-tower impact testing, and validated thermal management through CFD analysis. Analyzed 20+ samples.

Result

Achieved 32% weight reduction while improving impact energy absorption by 18%, enhanced ventilation airflow by 45%, passed all required safety certifications, and demonstrated 40% improvement in user comfort ratings during extended wear testing with 50 participants.

Innovation Highlight: The octet-lattice design principles developed in this project is a patent pending method currently being piloted by two major helmet manufacturers for their next-generation product lines, demonstrating real-world commercial impact.

Skills

- 3D Printing
- ANSYS
- FEA
- SolidWorks
- NTOPOLOGY
- Material Testing
- Failure Mode Analysis
- Rapid Prototyping
- Python



Applications of 3D Printed Octet Lattice Structures in Personal Protective Equipment

Ayodeji Abiodun • Matthew Bunce • Catherine Mubana • Bakul Mohan • Matthew Sparrow

Background

- Personal protective equipment (PPE) is a crucial part of preventing impact-related injuries in many facets of society, spanning from industry to sports (professional and recreational).
- There is a call for technological improvements in PPE to be pursued – with mass investments into PPE being made by the National Football League, and interest being shown in innovative helmet start-ups like HEDX®.
- Stereolithographic acrylate polymer resins in octet-truss conformations researched by Bolon and Bardelick are of potential use in PPE.

Approach

- Octet-truss structures of various thicknesses and strut lengths were explored to find the most suitable application via research conducted by Bolon and Bardelick.
- Optimal ethanol wash and ultraviolet curing times were considered following stereolithographic printing.
- Helmet cross-sections were created and tested using nTopology finite element analysis software, ignoring edge effects and stress concentrations for computational achievability.
- Compression tests were used to acquire stress-strain behavior.

Figure 1: Prototype Helmet with test-would cross-sections of varying octet-truss thickness

Figure 2: Static FEA Analysis of Helmet Cross-Section

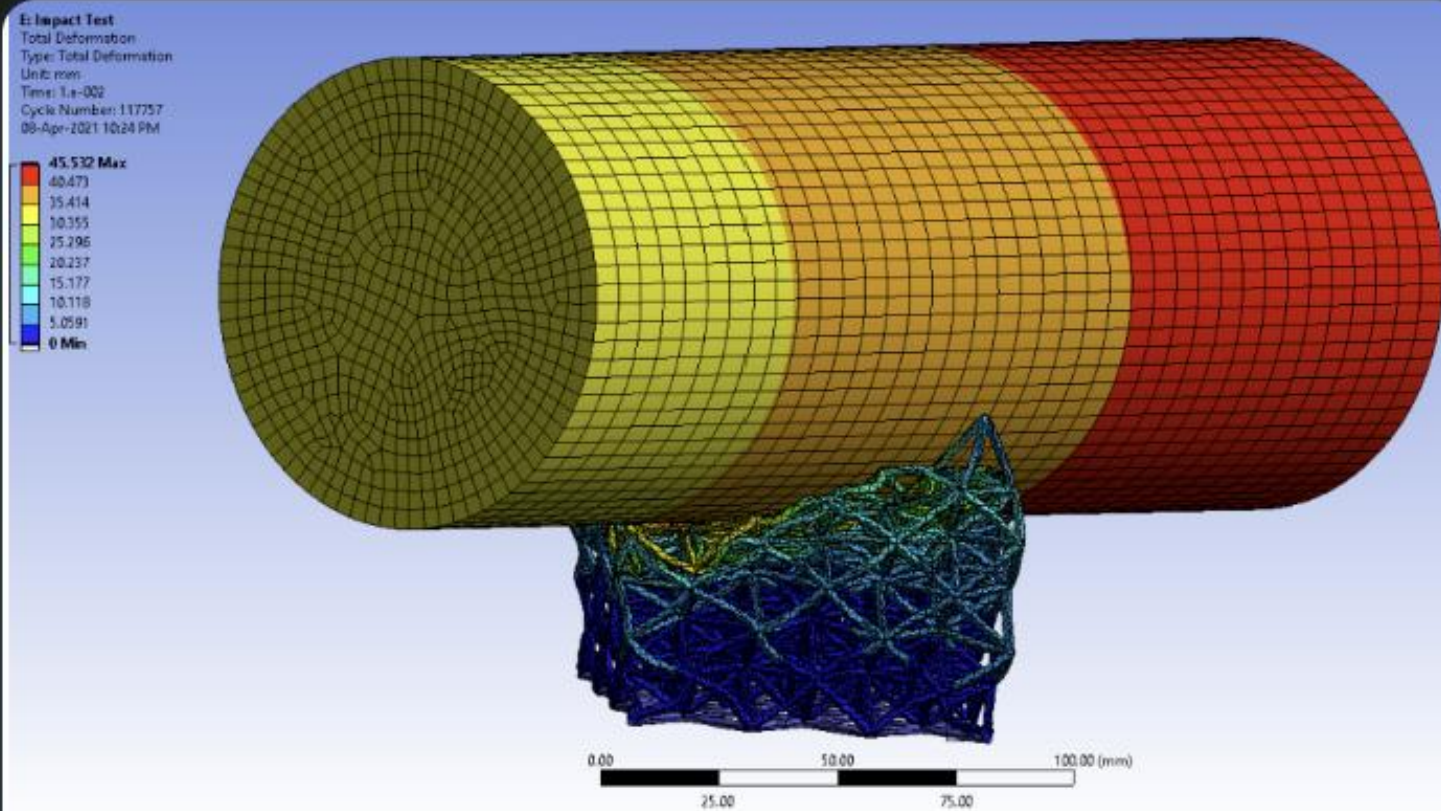
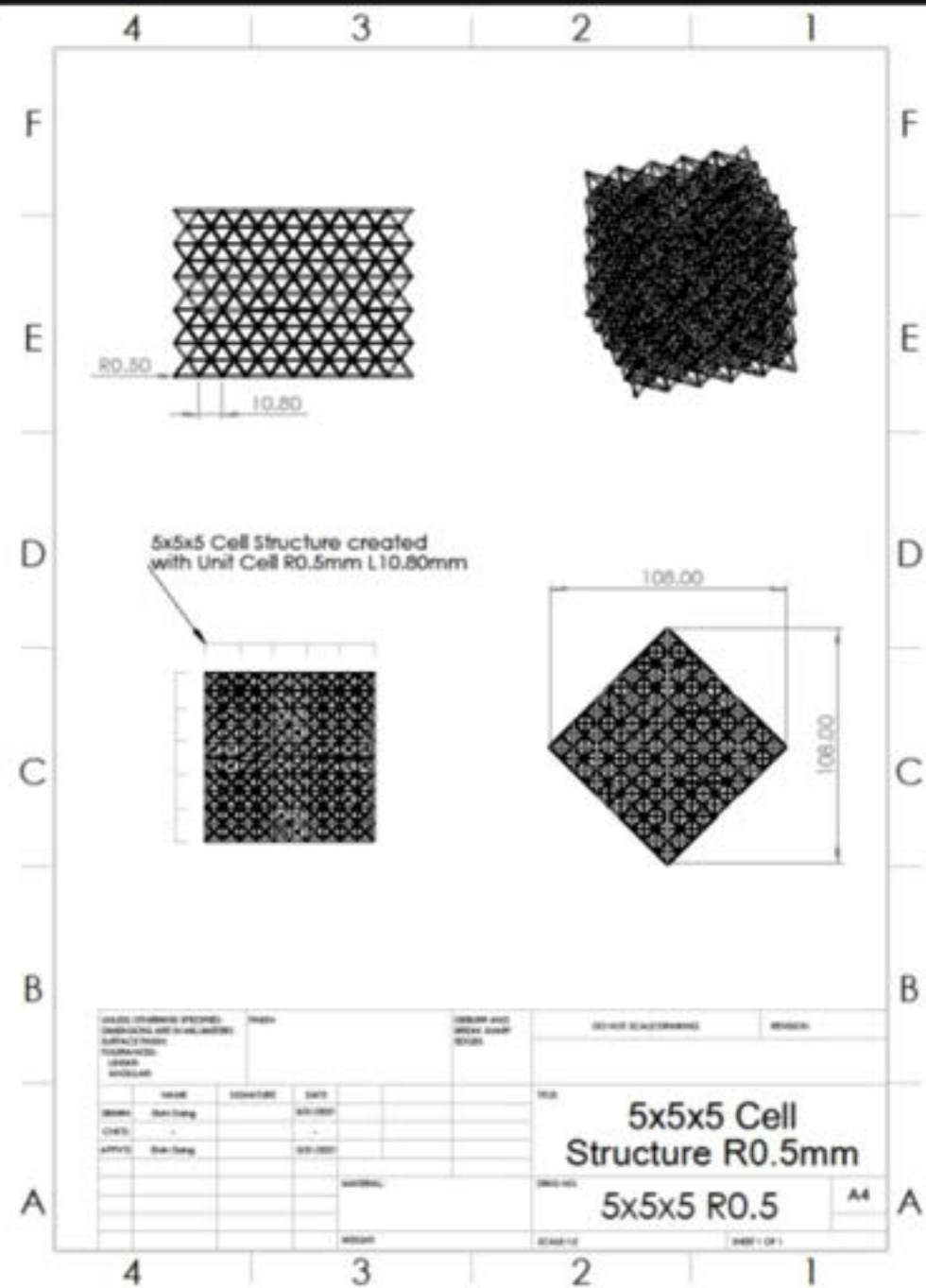
Figure 3: SLA-Printed Cross-Section Before (left) and During Curing (right)

Design

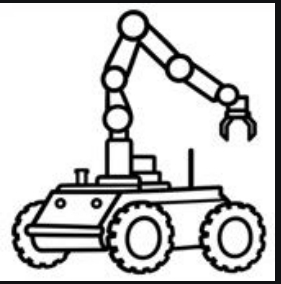
- Phrozen® Nylon-Green Tough Stereolithographic resin will be used, with a truss length of 9.0 mm and thickness of 1.50 mm.
- A solid inner layer was designed to reduce the dependency on supports, lowering cost and simplifying stereolithographic printing.

Conclusions

- The testing performed on the helmet cross-sections gives some evidence proving octet-truss structures printed using stereolithography has the potential for implementation into PPE.
- Advanced impact testing on a helmet prototype is needed, and MIPS technology should be integrated.
- Refine ergonomics and consider custom size printing.



Autonomous Mobile Manipulator for Data Center Rack Stacking



See Below

skills

- ROS2
- NAV2
- SLAM
MOVIT
RVIZ
- Network/Ethernet
- Failure Mode Analysis
- Rapid Prototyping

Autonomous Mobile Manipulator for Data Center Rack Stacking

Problem & Motivation

Data centers are expanding quickly, but server installation and replacement depend heavily on manual labor. **40%** of organizations suffered an outage caused by human error in the past 3 yrs, and **85%** are due to staff not following procedures [1].

What a successful run looks like:

Signal is sent to the husky (i.e Rack #3 shelf #2 unstack) -> husky autonomously navigates to rack -> image is scanned of the AprilTag (rack info and calibration) -> arm stack/unstacks -> Goes home

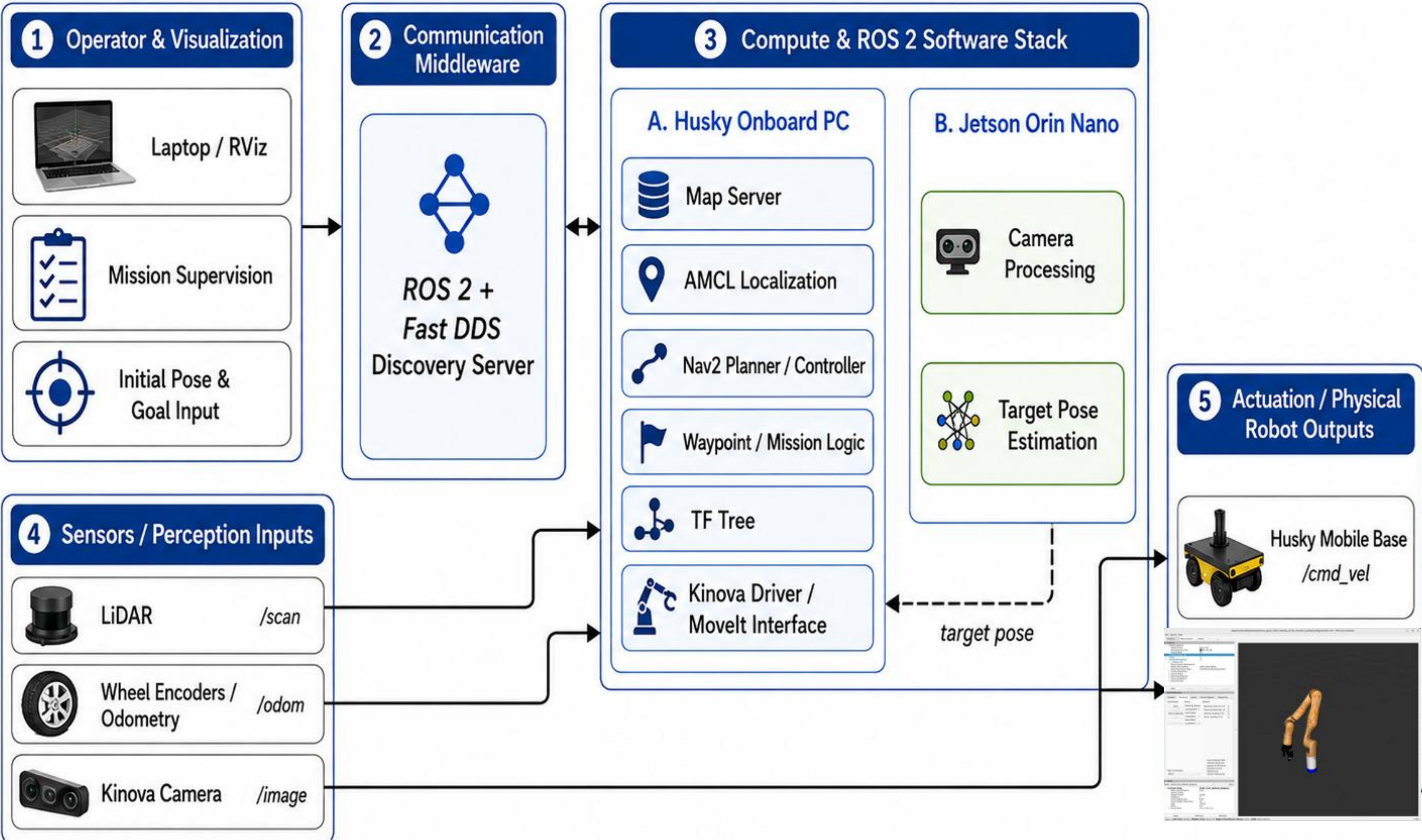
The Process from 0 to 100

Challenges solved:

- Network & connection
- Hardware wear/tear
- System integration
- Outdated software and documentation
- Camera Publisher connection



System Architecture



Applied Course Concepts

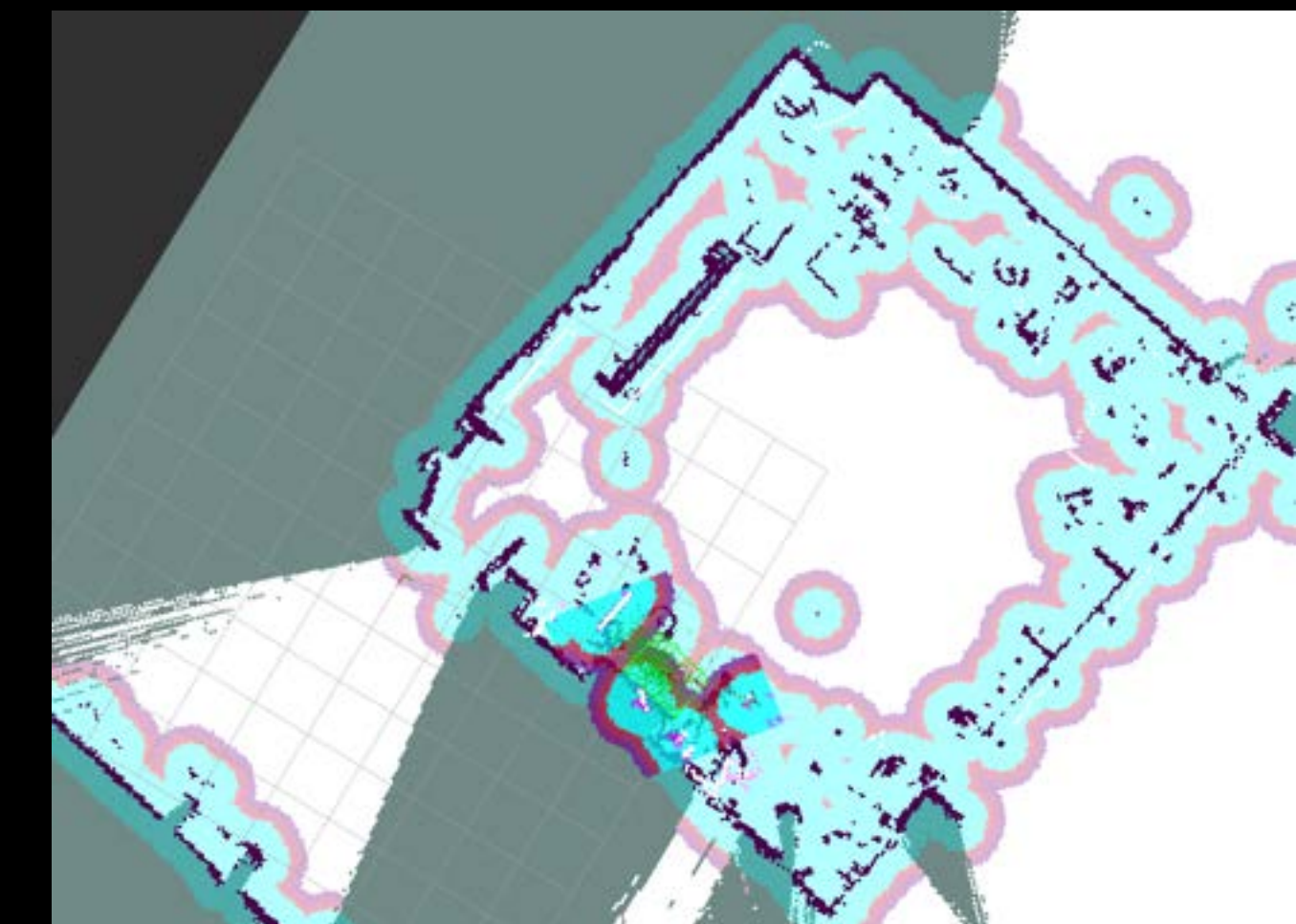
- Localization: AMCL is a particle-filter method matching LiDAR scans to map.
- Planning + control: separate global route planning from local trajectory tracking.
- Sensor fusion: odometry and joint states are proprioceptive; LiDAR and RGB camera are exteroceptive.
- Systems integration: lifecycle nodes, QoS and discovery.
- Frame transformations: Lidar, arm and camera to the base of the husky through ROS2.

Manipulation & Calibration

- AprilTag identifies the target rack and module ID.
- MoveIt 2 generates collision-free trajectories for the Kinova Gen3 manipulator.
- Predefined arm poses are manually calibrated to achieve repeatable grasping.
- Successful manipulation is verified by repeatable trajectory execution and stable grasping performance.

Navigation

- AMCL estimates $p(x,y,\theta)$ by comparing LiDAR scans to the map while odometry predicts motion.
- The global planner chooses the route; the local controller tracks it while costmaps block occupied cells.
- SLAM created a 2D occupancy-grid map of the racks, walls, and free space.



Evaluation & Results

Metric	Result
AprilTag Detection Rate	100% (Lab Tests)
Motion Planning Success	>95%
Navigation Success	>95%
ARM Start-up Time	~5–6 s
Grasp Success Rate	100%
Object Avoidance Success	85%

Table 1: Results for 15 trials run evaluating 6 metrics.

The system performed reliably in perception, motion planning, navigation, and grasping, but the lower 85% obstacle-avoidance success rate reveals the main weakness and safety risk because of the arm blocking it.

Progress

Weeks	Main Tasks	
1–2	Define the mock data-center rack plug-in scenario, confirm hardware availability, and set up ROS 2, robot model, coordinate frames, and basic Husky control.	✓
3–4	Set up sensors, including LiDAR and camera. Test RViz visualization, build or load a simple map, and prepare the mock rack and plug-in slot.	✓
5–6	Implement navigation to the target rack using Nav2. Add simple perception to detect the target rack or slot using a QR code, AprilTag, or colored marker.	✓
7–8	Set up Kinova arm motion planning and test low-speed plug-in motion on the mock rack. Integrate base alignment, target detection, and arm motion into one task sequence.	✓
9–10	Run experiments, collect evaluation data, and prepare the final demo, poster, presentation, and report.	✓ +

Table 2: Timeline and Milestone Progress.

Obstacle detection & Avoidance!



Future Work & Limitations

Future work could include improving on limitations:

- 5G or more stable WIFI to improve ROS 2 stability.
- Network latency reaches tens of ms, and the packet loss can be quite high as well.
- Lidar position and filter improvement.
- Navigation accuracy depends on map quality.
- Arm trajectories are not always optimal.
- Reliable hardware and updated software by Kionva and Clearpath.

WINNER - RescueVision: -See-Through-Walls-



About

Currently, many search and rescue operations are still a brute-force endeavor. In large-scale structural collapses, first responders can spend critical time clearing areas that are already empty simply because they lack the tools to verify signs of life from a distance. We realized the industry is still failing the people who risk everything to save others. Approach is inspired by:

- MIT CSAIL (Professor Dina Katabi): Her groundbreaking research on RF-Pose and wireless sensing demonstrated that radio frequency signals can be used to recover human silhouettes and track contactless vital signs.
- NASA JPL (FINDER Technology): NASA supported FINDER showed that the same signal processing principles used to track spacecraft could also detect the millimetric pulse of a human heart beneath rubble.

Task

create an AI-powered mobile navigation system providing real-time environmental analysis, obstacle detection, path guidance, and audio feedback specifically designed for search and rescue users. The solution needed to work offline, process video input in real-time, and provide accurate spatial awareness information.

Result

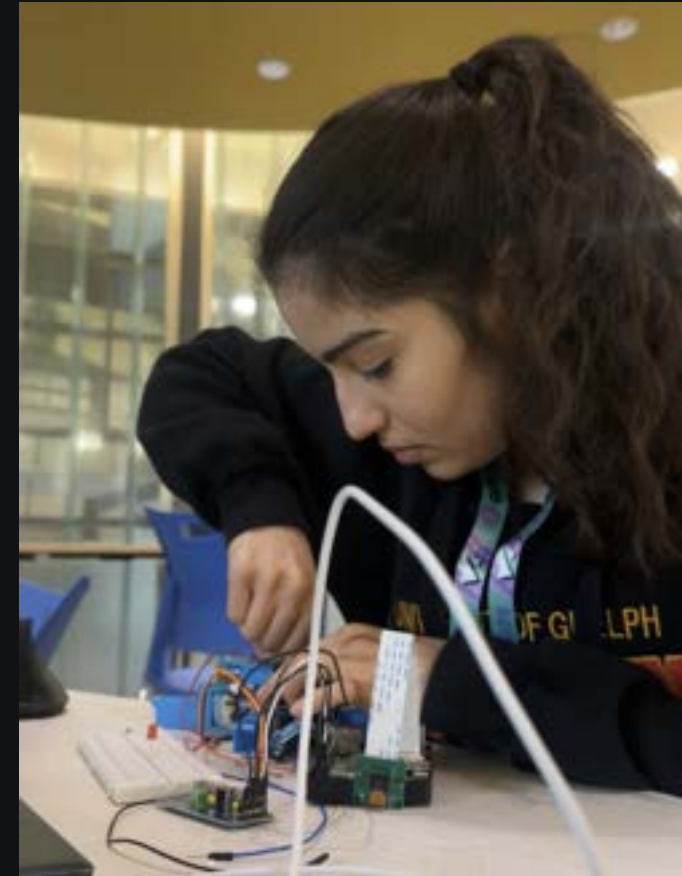
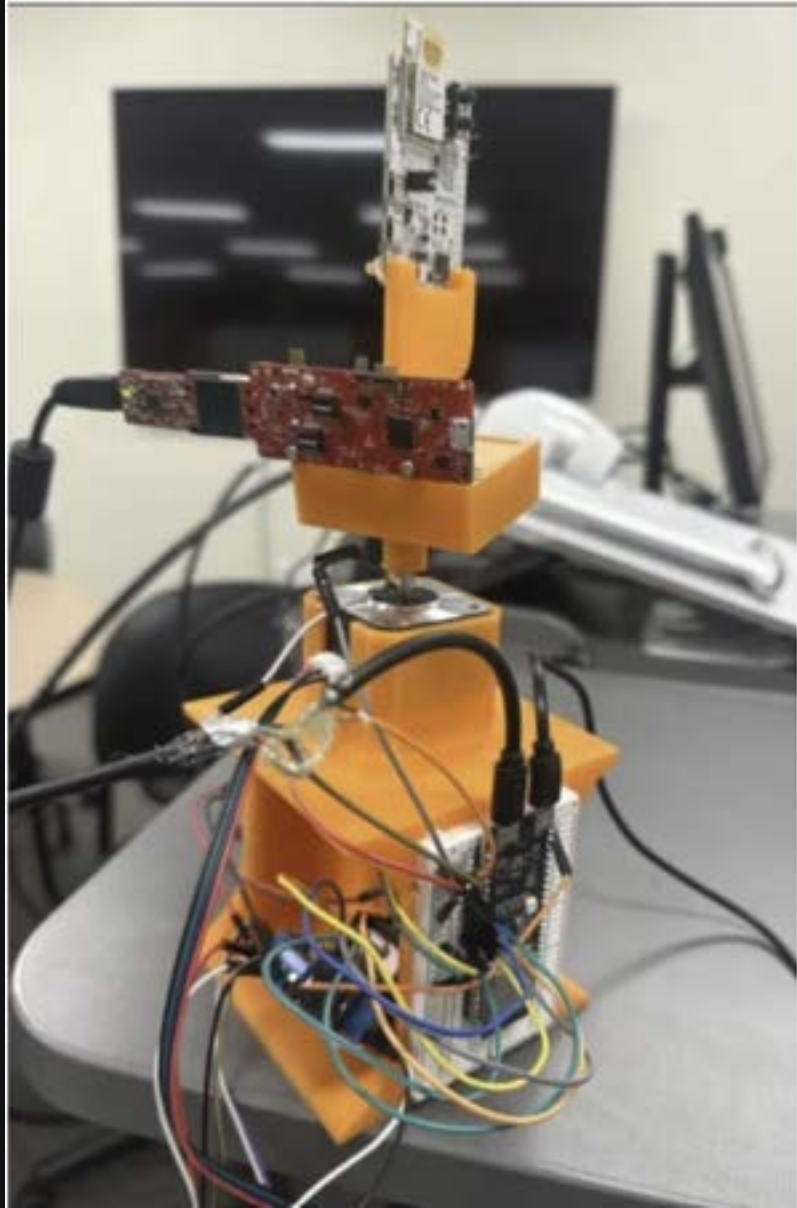
Won first place in with Analog Devices, achieved 94% accuracy in obstacle detection, reduced navigation time by average 35% during user trials, successfully guided 40+ test routes with zero safety incidents, Earned 4000\$.

RescueVision will be adapted with collaboration with texas instruments and Analog Devices!

Skills

- 3D Printing
- Edge AI Integration
- Time and Cost Management
- Rapid Prototyping
- Python
- AI Integration
- Sensor Fusion
- Teamwork

Devpost Link: <https://devpost.com/software/rescuevision-42j7ex>



Technical Skills Demonstrated Across Projects



Robotics & Automation

Advanced kinematics algorithms, servo motor control systems, PID feedback loops, motion planning, and real-time control implementation. Demonstrated through Q-Arm project with industrial-grade precision and reliability standards.



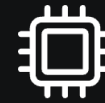
Machine Learning & AI

Computer vision algorithms, neural network implementation, gesture recognition, object detection, and adaptive learning systems. Applied in Visionify project with real-time processing and high-accuracy performance requirements.



Data Acquisition

LabVIEW programming, sensor calibration, measurement systems, statistical analysis, and automated testing protocols. Implemented in Tensile Tester project with precision measurement and standards compliance.



Embedded Systems

Custom PCB design, microcontroller programming, sensor integration, power management, and hardware-software interface development. Showcased in C-Wrist project with ultra-low-power optimization and real-time processing capabilities.



CAD & Simulation

SolidWorks 3D modeling, finite element analysis, topology optimization, stress simulation, and design validation. Utilized in Octet-Lattice helmet design with complex geometric optimization and safety compliance verification.



Software Development

Mobile app development, user interface design, cloud integration, API development, and cross-platform compatibility. Demonstrated across multiple projects with focus on user experience and performance optimization.

Each project in this portfolio represents a commitment to pushing boundaries while delivering solutions that make a meaningful difference in people's lives.
Thank you for viewing my passionate work!



[LinkedIn Profile](#)



[Portfolio](#)



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